

Hands-on Lab: Creating a Python Package



Creating a Python Package

Estimated time needed: **30** minutes

Objectives

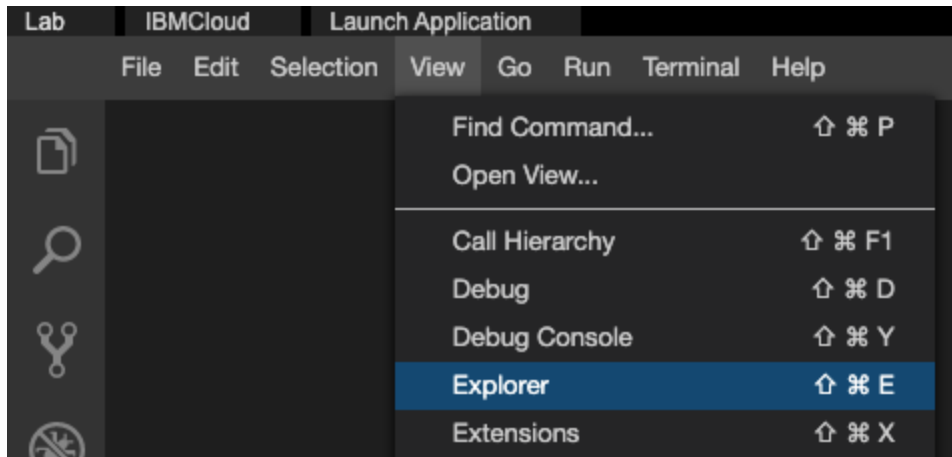
In this lab you will :

- Create a module named basic
- Add two functions to the module basic
- Create a module named stats
- Add two functions to the module stats
- Create a python package named mymath
- Verify that the package is working

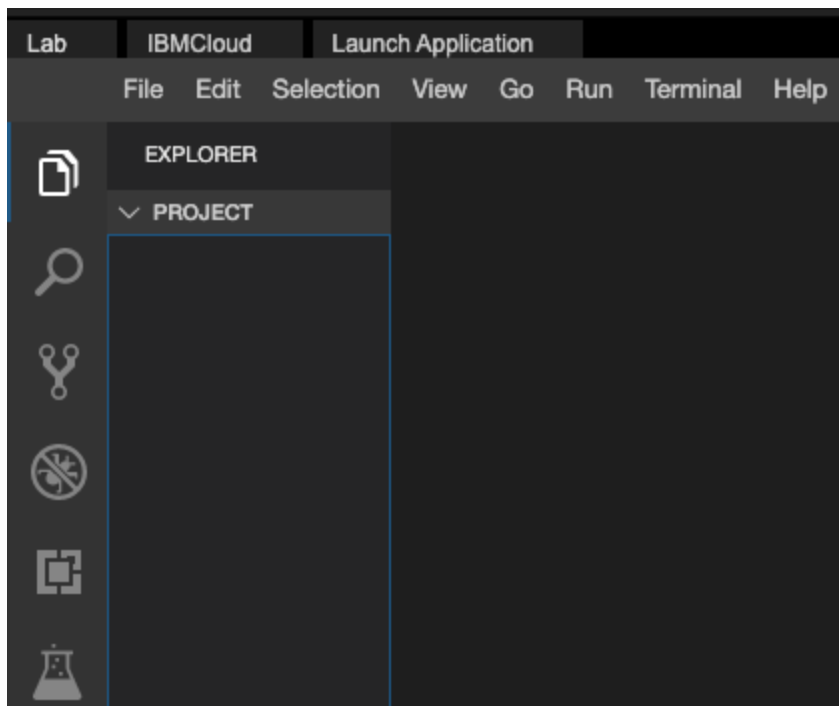
Lab

Create Package

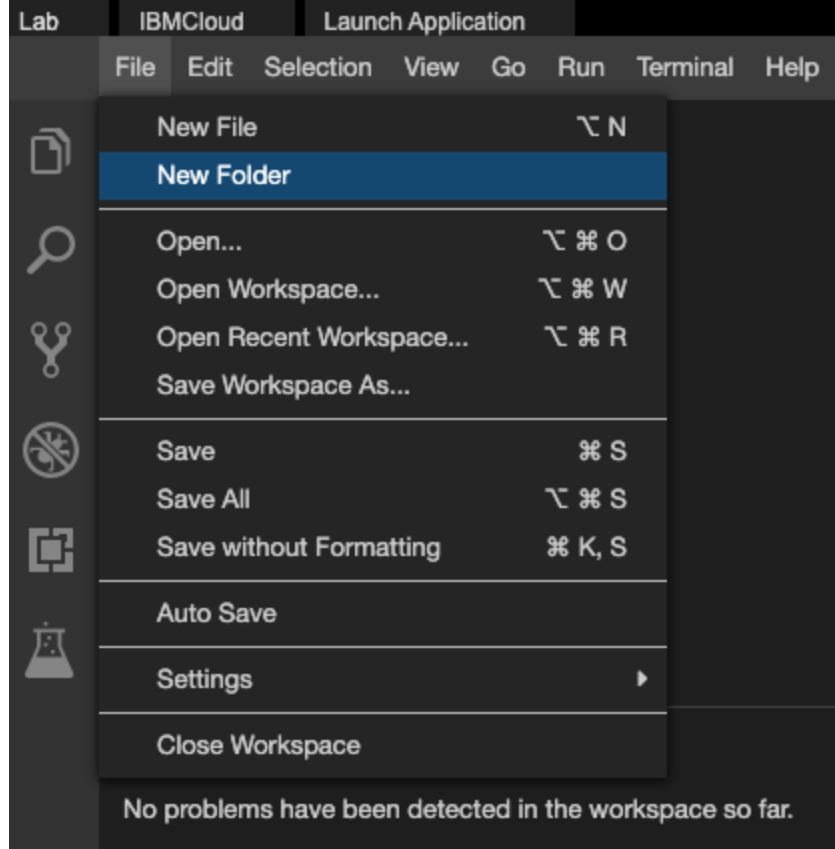
- On the window to the right, click on the **View** menu and select **Explorer** option, as shown in the image below.



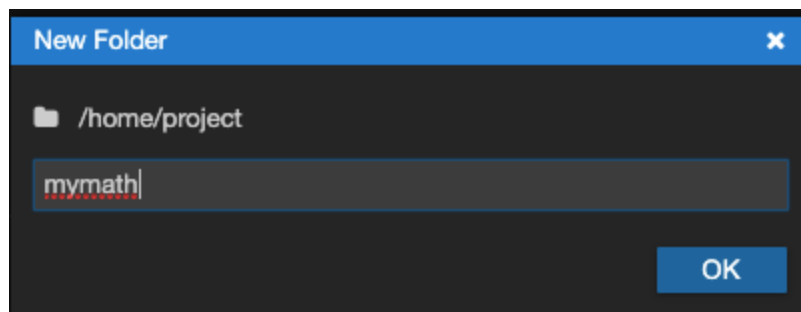
- Your IDE now should look like the image below.



- On the window to the right, click on the **File** menu and select **New Folder** option,



- Enter **mymath** and click OK as shown in the image below.

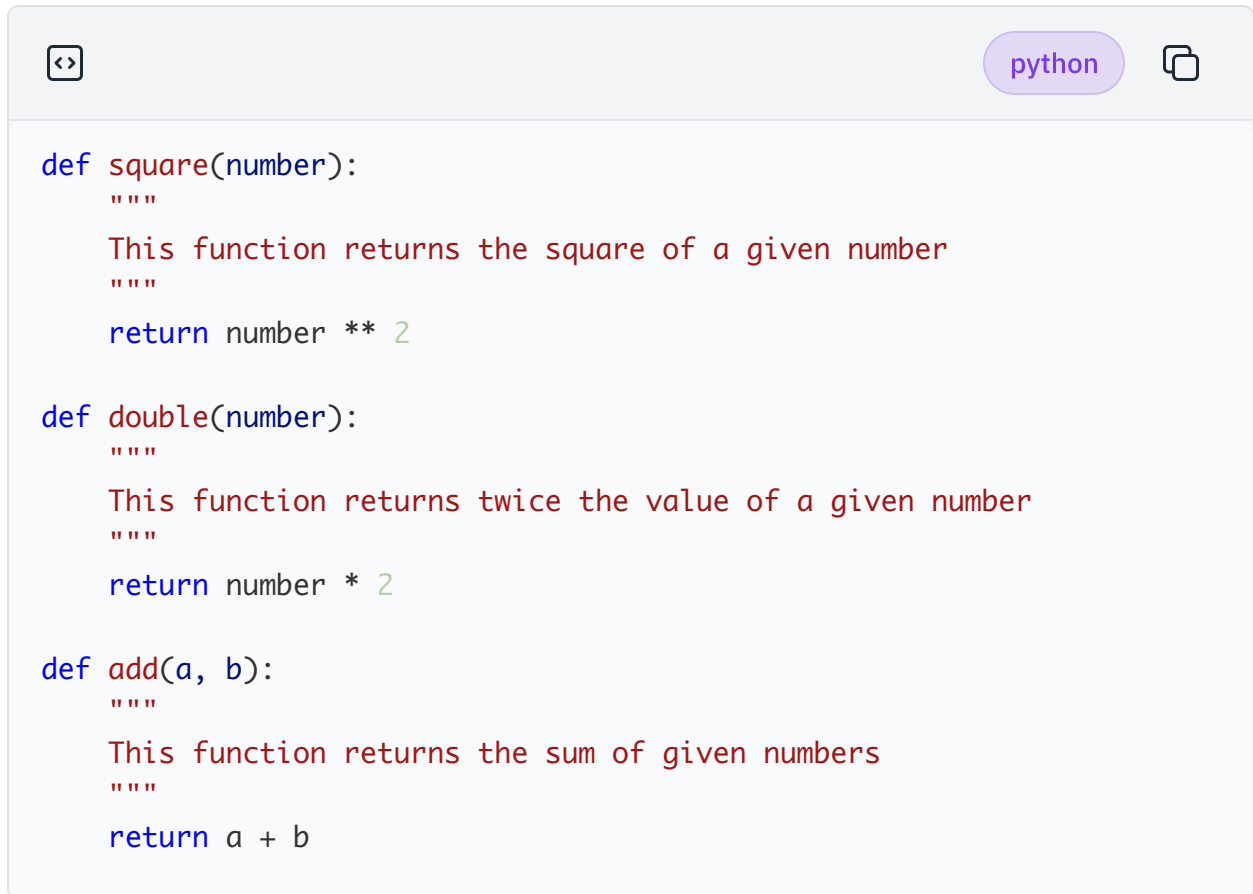


Create the first module

- Create a python module named basic

Create a file named **basic.py**.

Copy and paste the below code into basic.py

A code editor window with a light gray header bar. On the left is a code icon (two arrows pointing out of a box). On the right is a purple pill-shaped button with the text "python" and a copy icon (two overlapping squares). The main area is white and contains Python code with syntax highlighting: keywords are blue, strings are red, and numbers are green.

```
def square(number):  
    """  
    This function returns the square of a given number  
    """  
    return number ** 2  
  
def double(number):  
    """  
    This function returns twice the value of a given number  
    """  
    return number * 2  
  
def add(a, b):  
    """  
    This function returns the sum of given numbers  
    """  
    return a + b
```

You should see a screen like this now.

basic.py ●

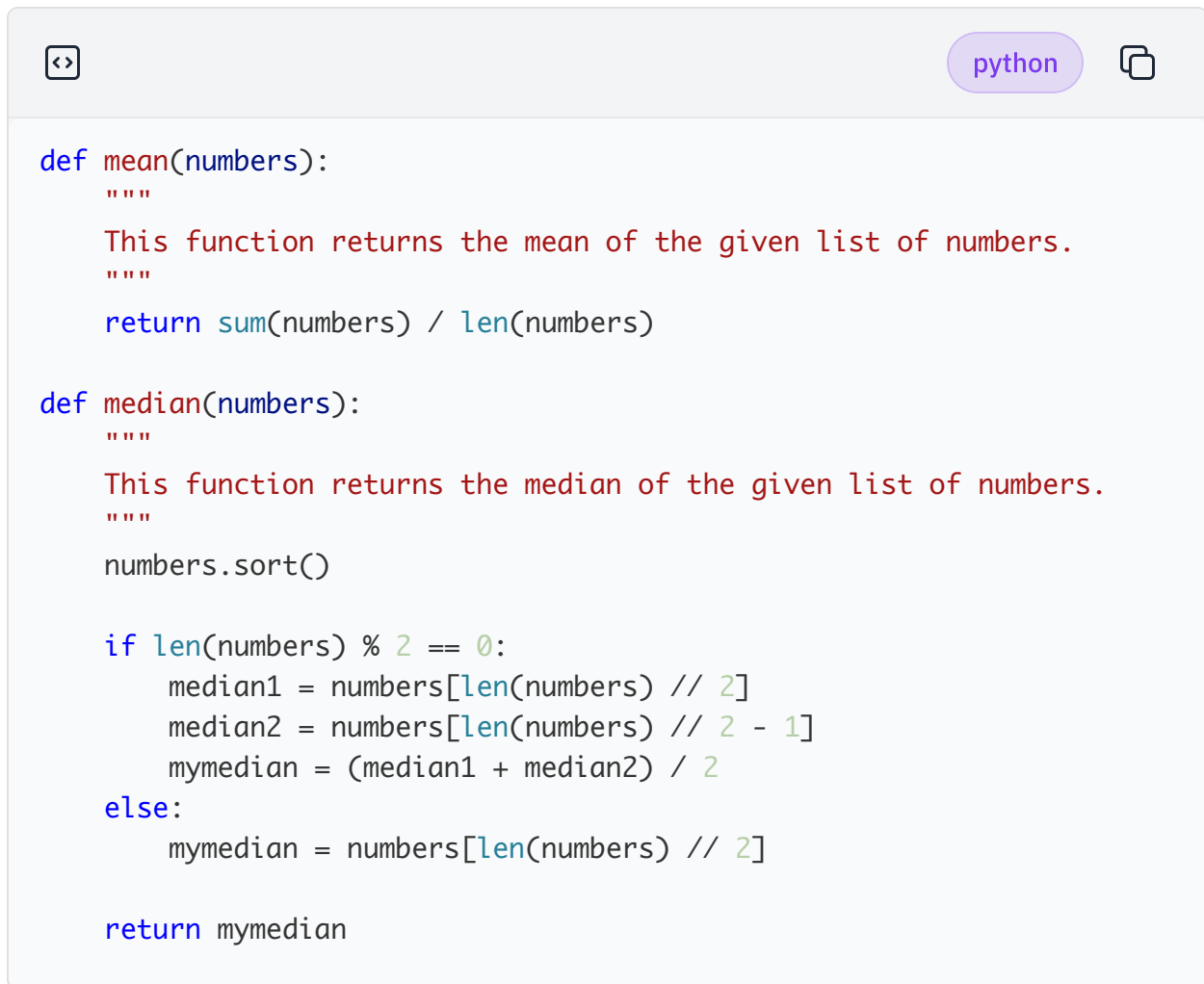
```
1  def square(number):
2      """
3      This function returns the square of a given number
4      """
5      return number ** 2
6
7
8  def double(number):
9      """
10     This function returns twice the value of a given number
11     """
12     return number * 2
13
14  def add(a, b):
15      """
16     This function returns the sum of given numbers
17     """
18     return a + b
19
```

Save the file **basic.py**

Create the second module

- Create a module named stats

Create a file named **stats.py** and copy and paste the below code into **stats.py**:



The image shows a code editor window with a light gray header. On the left is a code icon (<>), on the right is a 'python' label and a copy icon. The main area contains Python code for calculating mean and median of a list of numbers.

```
def mean(numbers):  
    """  
    This function returns the mean of the given list of numbers.  
    """  
    return sum(numbers) / len(numbers)  
  
def median(numbers):  
    """  
    This function returns the median of the given list of numbers.  
    """  
    numbers.sort()  
  
    if len(numbers) % 2 == 0:  
        median1 = numbers[len(numbers) // 2]  
        median2 = numbers[len(numbers) // 2 - 1]  
        mymedian = (median1 + median2) / 2  
    else:  
        mymedian = numbers[len(numbers) // 2]  
  
    return mymedian
```

You should see a screen like this now.

stats.py ●

```
1  def mean(numbers):
2      """
3      This function returns the mean of the given list of numbers
4      """
5      return sum(numbers)/len(numbers)
6
7
8  def median(numbers):
9      """
10     This function returns median of the given list of numbers
11     """
12     numbers.sort()
13
14     if len(numbers) % 2 == 0:
15         median1 = numbers[len(numbers) // 2]
16         median2 = numbers[len(numbers) // 2 - 1]
17         mymedian = (median1 + median2) / 2
18     else:
19         mymedian = numbers[len(numbers) // 2]
20     return mymedian
```

Save the file **stats.py**

Understanding the code

mean function: This function takes a list of numbers and returns their average. It uses Python's built-in **sum()** to add all values in the list and **len()** to get the count of elements, then divides the two to produce the mean.

median function: This function takes a list of numbers and returns the middle value. It begins by sorting the list in ascending order using **sort()**, since the median is position-dependent. It then checks whether the list has an even or odd number of elements. If the count is even, there is no single middle value, so it picks the two middle elements using index positions **len(numbers) // 2** and **len(numbers) // 2 - 1**, and returns their average. If the count is odd, there is a single middle element, and it is returned directly using index **len(numbers) // 2**.

Create `__init__.py`

- Create the file `__init__.py`

Copy and paste the below code into `__init__.py`

python

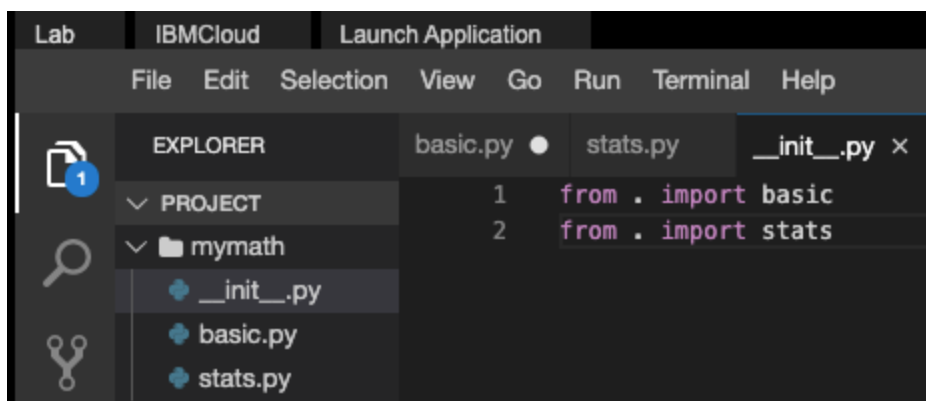
```
from . import basic
from . import stats
```

Save the file `__init__.py`

Now your directory structure should look like

plaintext

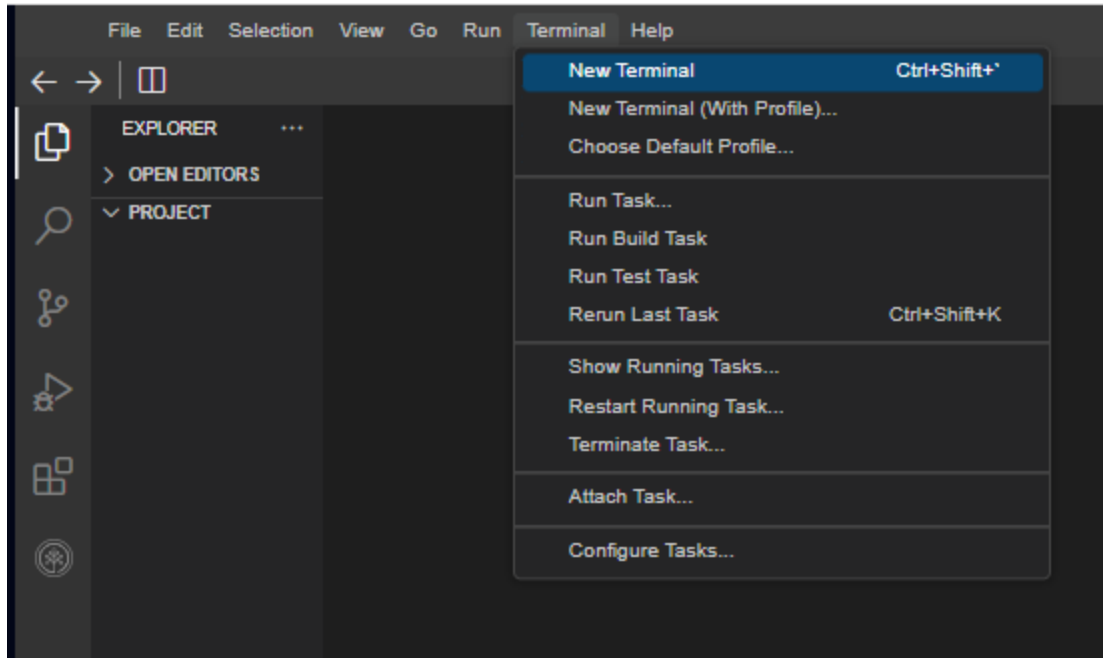
```
mymath
mymath/__init__.py
mymath/basic.py
mymath/stats.py
```



You are done creating a package

Verify the package

- On the window to the right, click on the **Terminal** menu and select **New Terminal** option, as shown in the image below.



- You will see a terminal open up on the bottom of the screen like the one in the image below.

```
theia@theiadocker-rsannareddy: /home/project ×  
theia@theiadocker-rsannareddy: /home/project$ █
```

- At the terminal type **python3** to invoke python interpreter.
- Once the python interpreter is loaded.
- At the python prompt type **import mymath**
- If the above command runs without errors, it is an indication that the mymath package is successfully loaded.
- At the python prompt type **mymath basic add(3 4)**

- You should see an output `7` on the screen.
- At the python prompt type `mymath.stats.mean([3,4,5])`
- You should see an output `4.0` on the screen.
- Type `exit()` to quit python interpreter.

```
theia@theiadocker-rsannareddy: /home/project ×  
theia@theiadocker-rsannareddy:/home/project$ python3  
Python 3.6.9 (default, Oct 8 2020, 12:12:24)  
[GCC 8.4.0] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>> import mymath  
>>> mymath.basic.add(3,4)  
7  
>>> mymath.stats.mean([3,4,5])  
4.0  
>>> exit()  
theia@theiadocker-rsannareddy:/home/project$ █
```

Practice Exercise

Create a new module named `geometry` and add to the `mymath` package.

- Create a module name `geometry`
- Add a function named `area_of_rectangle` that takes length and breadth as input and returns the area of a rectangle.
- Add a function named `area_of_circle` that takes radius as input and returns the area of a circle.
- Modify the `__init__.py` to include this module.
- Import and test the function `area_of_circle` from python terminal.

Authors

Ramesh Sannareddy

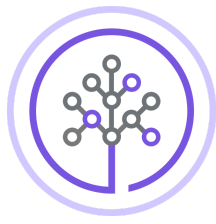
Other Contributors

Rav Ahuja

© IBM Corporation. All rights reserved.

This notebook and its source code are released under the terms of the [MIT License](#).

Package



Skills
Network

Creating a Python Package

Estimated time needed: **30** minutes

Objectives

In this lab you will :

- Create a module named basic
- Add two functions to the module basic
- Create a module named stats
- Add two functions to the module stats
- Create a python package named mymath
- Verify that the package is working

Lab →